

E-Commerce Supply Chain Analytics SQL Project

10 Business-Critical SQL Questions

Dataset: E-Commerce Order Fulfillment (100K Records)

LEVEL 1: FOUNDATIONAL OPERATIONAL METRICS (EASY)

Q1. Overall On-Time Delivery & Late Delivery Rate

EASY

Write a query to calculate total orders, count of late deliveries, and the overall late delivery percentage across all fulfillment channels.

SQL Skills: COUNT(), SUM(CASE WHEN), Percentage Calculation

Business Intent: Establishes baseline Service Level Agreement (SLA) compliance and global delivery performance.

KPI Impact: Measures On-Time In-Full (OTIF) baseline metrics for executive dashboarding.

Q2. Product Category Delay Profiling

EASY

Aggregate total order volume and count of delayed shipments (`late_delivery = 1`) grouped by product category. Sort by highest delayed volume.

SQL Skills: GROUP BY, ORDER BY, Categorical Aggregation

Business Intent: Identifies merchandise categories subject to systemic fulfillment or packaging delays.

KPI Impact: Pinpoints high-risk product lines to optimize warehouse pick-and-pack strategy.

Q3. Top 5 Geographical Delivery Bottlenecks

EASY

Retrieve the top 5 customer cities experiencing the highest absolute number of late deliveries.

SQL Skills: WHERE, GROUP BY, LIMIT / TOP

Business Intent: Isolates regional delivery hotspots requiring carrier renegotiation or local 3PL hub deployment.

KPI Impact: Highlights regional logjams to re-route last-mile distribution corridors.

Q4. Warehouse Dispatch Delay vs. Priority Level

EASY

Calculate the average dispatch delay in hours (`dispatch_delay_hours`) broken down by order priority level.

SQL Skills: `AVG()`, `GROUP BY`, Metric Benchmark

Business Intent: Validates whether warehouse fulfillment queues successfully prioritize High/Critical orders over Standard orders.

KPI Impact: Evaluates internal warehouse staging and order pick queuing efficiency.

LEVEL 2: VENDOR & ENVIRONMENTAL DIAGNOSTICS (MEDIUM)

Q5. Seller Reliability Scorecard & SLA Filtering

MEDIUM

For sellers with at least 50 total orders, aggregate average seller rating, average reliability score, and late delivery rate. Filter for low-performing sellers (reliability score < 3.0 and late rate > 25%).

SQL Skills: `HAVING` clause, Conditional Aggregation, Vendor Auditing

Business Intent: Enforces vendor performance management and penalizes sub-par seller fulfillment practices.

KPI Impact: Mitigates supply network risk by identifying non-compliant third-party merchants.

Q6. External Friction Analysis (Weather & Traffic)

MEDIUM

Group shipments by combined weather condition and traffic congestion level. Compute total shipment count, late delivery rate, and mean transit delay.

SQL Skills: Multi-column `GROUP BY`, Ratio Calculations, Environmental Analysis

Business Intent: Quantifies unavoidable external environmental delays versus controllable internal operational delays.

KPI Impact: Refines dynamic Promised Delivery Date (PDD) algorithms based on real-time transit risk profiles.

Q7. Distance Tiering & Dispatch Latency

MEDIUM

Bin warehouse distances (`warehouse_distance_km`) into tiers ('<50 km', '50-200 km', '>200 km') and compare dispatch delays and failure rates across each bin.

SQL Skills: `CASE WHEN` Binning, Aggregate Functions, Logistical Segmentation

Business Intent: Determines optimal inventory positioning and fulfillment center allocation relative to customer density.

KPI Impact: Informs network expansion and strategic placement of regional fulfillment centers.

LEVEL 3: STRATEGIC OPTIMIZATION & RISK MODELING (HARD)

Q8. Intra-State Category Performance Ranking

HARD

Rank product categories within each customer state based on late delivery percentage using window functions. Filter to output only the single worst-performing category per state.

SQL Skills: `DENSE_RANK()` `OVER (PARTITION BY)`, CTEs / Subqueries

Business Intent: Allows localized supply chain leads to tailor state-level carrier selection and inventory buffers.

KPI Impact: Enables localized root-cause targeting for state-specific fulfillment bottlenecks.

Q9. High-Value Priority SLA Breach & Revenue Risk

HARD

Isolate high-value orders (> \$200) with 'High' or 'Critical' priority that breached SLAs. Compute total revenue at risk per state and calculate cumulative running totals.

SQL Skills: `SUM()` `OVER (ORDER BY)`, Complex Multi-Filter CTEs

Business Intent: Focuses executive attention on high-value client churn risks resulting from delivery failures.

KPI Impact: Protects high-margin account revenue by establishing emergency expediting protocols.

Q10. Multi-Factor Root-Cause Bottleneck Profiling

HARD

Identify compound risk scenarios (e.g., Low Seller Reliability + Severe Weather + High Distance) where late delivery rates exceed 40%. Compare warehouse delay variance vs. carrier transit variance.

SQL Skills: Compound CTEs, Advanced Variance/Standard Deviation functions, Statistical Grouping

Business Intent: Builds a comprehensive predictive risk matrix to trigger proactive dynamic re-routing.

KPI Impact: Reduces tail-end delivery risk through proactive order re-assignment and predictive alerting.